

Project Design Phase

Problem - Solution Fit Template

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Problem - Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- ✓ Solve complex problems in a way that fits the state of your customers.
- ✓ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ✓ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ✓ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ✓ Understand the existing situation in order to improve it for your target group.

Template:

1. CUSTOMER SEGMENT(S) CS Who is your customer? Independent/self-paced online learners across academic fields (Computer Science, Mathematics, Physics, Chemistry, Biology, Engineering, Business, Literature, History, Psychology) studying without a live instructor present.	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? Limited time between study sessions; hesitation/embarrassment to ask a human tutor for help; no 24/7 access to a live tutor; budget constraints for private tutoring; wanting a quick answer, not a full course.	5. AVAILABLE SOLUTIONS AS Which solutions are available to customers when they face the problem, or to get the job done? Re-reading material repeatedly; searching YouTube/forums (Reddit, Discord study groups); generic FAQ chatbots with no emotional awareness; emailing a TA and waiting; silently giving up.
2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one. Get unstuck when confused; regain motivation when bored; get validated and challenged further when confident; get deeper material when curious; de-escalate frustration after repeated failed attempts.	9. PROBLEM ROOT CAUSE RC What is the real reason this problem exists? What is the back story behind the need to do this job? Free-text learning interactions carry emotional signal (confusion, frustration, boredom, curiosity, confidence) that platforms never capture or act on - a confused student and a confident student receive an identical, generic response.	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? Directly or indirectly related. Closes the tab/app when stuck instead of asking for help; re-attempts the same problem with the same approach; skips ahead without understanding prior material; searches external sources when the platform doesn't help.

3. TRIGGERS TR What triggers customers to act? e.g. seeing a neighbour try a new solution, reading about it in the news. Repeatedly failing the same problem; an approaching assignment deadline; comparing progress to peers who seem to understand faster; seeing a relatable quick-example prompt in the app.	10. YOUR SOLUTION SL If you are working on an existing business, write your current solution fit. If new, keep it blank until filled. A Streamlit app where a student types their field and learning problem in plain text. Two independent models (a BiLSTM and a fine-tuned BERT) classify the emotional state behind the text across 5 classes (Bored, Confident, Confused, Curious, Frustrated), including detecting mixed emotions. A keyword-boosting layer sharpens ambiguous cases, and the detected emotion plus confidence score are passed to the Gemini API to generate a personalized, field-specific, emotionally appropriate response - with a template fallback if the API is unavailable. All interactions are logged for future analysis, and an analytics dashboard shows the student's emotional trend across the session.	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Search engines, YouTube tutorials, Reddit/Discord study groups, course platform's own comment section, emailing a TA/LMS messaging. 8.2 OFFLINE What kind of actions do customers take offline? Asking classmates in person, attending office hours, in-person study groups, calling/texting a tutor or parent for help.
4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or job, and afterwards? Use in communication strategy & design. Before: frustrated, anxious, embarrassed to ask, disengaged. After (with this solution): supported, validated, encouraged, with a clearer concrete next step instead of feeling alone with the		

problem.

References:

1. <https://www.idealhackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>